

# Closure-Validated Circuit Discovery in Attention Heads: Co-activation Proposes, Ablation Disposes

Abby Xu

Working draft, May 2026

## Abstract

A growing body of interpretability work treats *groups* of features, not individual units, as the basic unit of representation. A natural methodological corollary is to discover such groups by clustering feature co-activation statistics. We study a specific instance of this idea: *attention-head* circuits in 1B-scale transformer language models. Our object and validation differ from work on sparse-autoencoder feature manifolds (e.g., Bhalla et al. [2], which we draw the clustering recipe from but do not evaluate on its own terrain): we cluster attention heads rather than SAE features, and we validate by causal ablation rather than by manifold reconstruction. The question we ask is whether co-activation clusters of attention heads are *circuits* — load-bearing subsystems whose removal degrades the model’s behavior — or merely communities of heads that fire together for reasons unrelated to function. Without specifying any target template, we cluster head-focus statistics via a pairwise Ising model and then run a closure test: ablate the discovered community, compare per-example damage to five matched-random-head controls, and report multi-metric specificity (cross-entropy loss, top-1 accuracy, mean target-token logit). Across two dense 1B-scale models, OLMo 1B and Pythia 1B, and two input distributions, synthetic induction and natural text, all four dense closure tests pass with the candidate’s damage in the predicted direction and well outside the random-control distribution on at least one metric ( $z$ -scores from  $+1.83\sigma$  to  $+36.4\sigma$  on loss, up to  $-52.5\sigma$  on target-logit). One result, Pythia 1B on natural text, shows a  $25\times$  divergence between metrics that we read as a *downstream-redundancy signature*: a real specific circuit whose output is partially reconstructed by later layers, so the aggregate loss barely notices what the per-target logit reports as a  $52\sigma$  effect. The MoE setting (OLMoE-1B-7B) tells a more cautionary story. The marginal Ising collapses on natural text (ARI 0.006 vs 0.193 synthetic). Route-conditional stratification of examples by per-layer routing weights recovers a  $+3.05\sigma$  statistical signal against a random-partition null — but the recovered community does not pass closure: ablating it *improves* loss, more so on inputs outside the discovered route stratum than inside, in the direction opposite to a route-conditional circuit. Finally, extending the same closure test across training checkpoints, we find that two other cheap proxies — attention-to-target selectivity and participation ratio — also fail to track closure-validated function: BOS-head function is load-bearing  $\approx 2B$  tokens before its attention pattern forms (function without form, in two dense architectures), while previous-token heads can show a sharp attention pattern and high participation ratio with no closure signal (form without function). We conclude that, for attention-head circuits, a cheap signal — cluster membership, attention selectivity, or participation ratio — is a circuit *proposal* rather than a confirmed circuit: closure testing is what separates the two, it is necessary in the dense models we study and not yet sufficient in the MoE setting even after route conditioning, and multi-metric closure reporting (target-logit and accuracy as primary indicators, loss as a conservative floor) is the right way to read tests of this form.

# 1 Introduction

Mechanistic interpretability has converged, in recent years, on an ensemble-of-units view of how neural networks compute. Sparse autoencoders find that concepts are tiled across many features rather than localized to one [2]; attention behaviors are distributed across multiple heads rather than implemented by a canonical single head [5]; continuous concepts (color, day-of-week, age) traverse low-dimensional manifolds in mid-network activations [4, 9]. Across these results, the interpretive unit is a group — a subspace, a manifold, a head-set — and not a direction.

A natural methodological move follows: if the interpretive unit is a group, find groups by clustering. Co-activation statistics across a fixed batch of inputs become the data; affinity between two units (feature similarity, mutual information of binary activations, or conditional couplings in a pairwise Ising model) becomes the proposed edge weight; spectral clustering or modularity optimization on the resulting affinity graph yields candidate groups. Bhalla et al. [2] formalize a version of this for SAE features and recover manifold-tiled groups whose joint activity tracks the underlying geometry.

The question this paper asks is not whether clustering finds *something* — it clearly does — but whether what it finds is a **circuit** in the load-bearing sense of mechanistic interpretability: a group of components whose removal degrades the model’s behavior on inputs where the group’s putative function applies. The distinction matters because the same clustering output is consistent with several different underlying realities:

1. A group of heads that jointly implement a function, whose removal substantially damages the function on relevant inputs (a *circuit*);
2. A group of heads that fire together because they share an upstream driver (a shared input, a routing pattern, a positional bias) but contribute redundantly or marginally to the output (a *co-active community* that is not a circuit);
3. A group of heads that share a co-activation pattern because they all add a similar bias signal that the rest of the network compensates for at the output (a *cancellable community* whose removal might even help).

The clustering output alone cannot distinguish these. A test that *can* distinguish them is straightforward: ablate the discovered community by zeroing the per-head outputs that contribute to the residual stream, compare per-example damage on the same batch to that produced by matched random-head ablations, and ask whether the candidate’s damage is (a) in the direction expected of a real circuit (loss up, accuracy down, target-token logit down) and (b) outside the distribution of effects produced by random head-sets of the same size.

We run this contrast on three 1B-scale language models — Pythia 1B, OLMo 1B, OLMoE-1B-7B — and two input distributions — a synthetic induction batch and a natural-text batch from prior probe-circuit work [12]. The dense models, on both distributions, produce co-activation communities that pass closure: the candidate’s ablation damage is in the predicted direction and significantly above the random-control distribution. The MoE model, on natural text, does not. Its marginal Ising signal collapses entirely. Route-conditional stratification — clustering examples by their routing pattern and fitting separate Isings within each route stratum — recovers a statistical signal that survives a careful random-partition null control, but the recovered community then fails closure in the wrong direction: ablation improves loss, more so on inputs outside the stratum the community was discovered on.

We treat this asymmetry as the paper’s main finding. The dense closures show what clustering-plus-closure can do; the MoE closure shows what clustering-plus-route-conditioning-plus-closure cannot do, in a case where the clustering looks methodologically careful and the statistical signal

looks real. The gap between proposal and discovery is architecturally specific, and it widens to wrong-direction in MoE on natural inputs even after the obvious fix.

We then ask the same question of two other cheap proxies, across the training axis. Using cached intermediate checkpoints, we measure when attention-to-target selectivity and participation ratio emerge for each head class and when the class becomes load-bearing under closure. The two do not coincide: BOS-head function is load-bearing  $\approx 2B$  tokens before its attention pattern forms in two dense architectures (function without form), while previous-token heads at one checkpoint show a sharp attention pattern and high participation ratio with no closure signal (form without function). The bidirectional crossing shows attention pattern and function are distinct constructs, not the same construct at different sensitivities — the same lesson as the co-activation result, now for two more proxies.

We also document a multi-metric reading of closure that we recommend as a standard. Cross-entropy loss, taken alone, would misrank our community-closure tests: it inflates one positive result by control-variance collapse (OLMo 1B natural,  $z = +36\sigma$ ), understates another by metric-aggregation slack (Pythia 1B natural,  $z = +1.95\sigma$  on loss,  $z = -52.5\sigma$  on target-logit), and gives a numerically large but qualitatively wrong-direction value for the MoE failure. Target-token logit and top-1 accuracy each give a more stable ranking and combine to give a coherent verdict. The Pythia 1B natural-text divergence between metrics is itself an architectural finding: a  $25\times$  gap in  $z$ -scores within the same test on the same ablation. We read it as a downstream-redundancy signature — the discovered circuit is real and specific, but the model has redundant downstream pathways that partially reconstruct the ablated community’s output, so the loss aggregate is partially insulated even while the per-target logit collapses.

The paper exhibits five community-closure tests as its primary evidence base (§4–§5, summarized in §6), extends the closure test across training to two further proxies (§7), and collects its claims and an explicit *what we do not claim* scope statement in §8 — the temptation to overgeneralize from this evidence is real and we mark out the scope carefully.

## 2 Related work

**Co-activation and group-level interpretability.** Bhalla et al. [2] develop a framework in which sparse-autoencoder features are grouped post-hoc by conditional co-activation, fitting a pairwise Ising model on binarized SAE codes and recovering feature communities that align with continuous concept manifolds (age, color, day-of-week, temperature). Their pipeline is the most direct methodological analog to what we test here, and we adapt their binarization-plus-Ising recipe to attention-head focus statistics. Two important differences from their setting: (i) we apply the recipe to a discrete-circuit domain (attention heads with classifiable capability functions — induction, previous-token, first-token, etc.) rather than continuous concept manifolds; (ii) we ablate the discovered communities to test their causal load-bearing-ness, whereas Bhalla et al. validate communities by reconstruction coherence (whether atoms grouped together collectively reconstruct the manifold).

**Distributed attention behaviors.** Goodfire’s adversarial parameter decomposition [5] decomposes weight matrices into rank-1 subcomponents and observes that single attention behaviors (previous-token, syntax-boundary) distribute across multiple heads rather than localizing. This is the parameter-side counterpart to what clustering observes on the activation side: real attention functions inhabit subspaces, not single heads.

**Linear vs. nonlinear representations.** The Linear Representation Hypothesis [9, 8] treats concepts as one-dimensional directions in activation space; subsequent work [4, 6] finds that this is insufficient for many concepts (circular topologies for periodic concepts; polar coordinates for syntactic structure). Our setting is discrete-concept (attention head classification by capability), so LRH-vs-manifold isn’t directly the issue — but the underlying philosophical move, that interpretation lives in subspaces rather than single directions, is shared.

**Circuit discovery and validation.** The mechanistic-interpretability literature on circuits typically combines (i) a hypothesis about a specific function (induction, IOI, indirect-object identification), (ii) a probe or attribution graph that identifies candidate components, and (iii) ablation or path-patching to validate the components’ causal role [7, 11, 3]. The novel methodological question we test is whether step (ii) can be replaced — partially or wholly — by *unsupervised* co-activation clustering with no function specified in advance, with step (iii) serving as the validation gate. The dense results suggest this is feasible for some functions in dense models; the MoE result shows where the substitution breaks down.

**MoE interpretability.** Mixture-of-Experts models have been studied primarily through the experts themselves — what each expert specializes in, how routing patterns correlate with input type [13]. The attention-head circuits of MoE transformers are much less studied, and our finding that route-conditional clustering recovers a statistical signal that fails closure is, to our knowledge, the first systematic comparison of co-activation interpretability between dense and MoE architectures.

## 3 Method

### 3.1 Overall design

The study tests four claims (made precise in §8) by running five closure tests. Each closure test consists of three stages: (i) *proposal* — given a model and a batch, cluster attention heads by co-activation statistics and select the cleanest candidate community; (ii) *prediction* — declare the candidate a circuit if ablation damages the model’s behavior in a predicted direction; (iii) *closure* — ablate and compare per-example damage to five matched random-head controls, reporting multi-metric  $z$ -scores. The four dense  $\times$  two-distribution tests cover *whether the dense pipeline finds circuits across model and input distribution*. The fifth (MoE  $\times$  natural-text  $\times$  route-conditional) covers *whether the obvious extension to MoE recovers circuits*.

### 3.2 Forward extraction

For each model and input batch, we run a forward pass with `output_attentions=True` and extract the per-(example, layer, head) attention pattern at a per-example *query position*. The query position is the last token for the synthetic batch and a per-example index (saved with the batch from prior work) for the natural batch. The extracted tensor has shape  $(N, L, H, T)$  where  $N = 2000$  examples,  $L$  is the number of layers,  $H$  is the number of heads per layer, and  $T = 256$  is the sequence length.

For the MoE forward pass we additionally set `output_router_logits=True` and extract the softmax routing weights at the query position per layer, giving a tensor of shape  $(N, L, E)$  where  $E = 64$  is the number of experts per layer (OLMoE).

### 3.3 Template-free signal

For each example  $i$  and head  $(L, H)$ , we collapse the attention pattern at the query position to a single scalar:

$$\text{sig}_{i,(L,H)} = \max_k \text{attn}_{i,L,H,k}.$$

This is the maximum attention weight assigned by the head to any key position in this input. High values indicate that the head is *focused* on something (some particular key); low values indicate a near-uniform distribution over keys. The signal is **template-free** in the sense that no specific key position (BOS, previous-token, induction target, etc.) is privileged — only the head’s degree of focus matters.

This choice mirrors Bhalla et al.’s use of SAE code magnitudes as their template-free signal. Alternatives we considered and did not use: attention entropy (low entropy = focused; equivalent up to monotone transform to max for most heads), attention to a specific canonical target (e.g., BOS — but this hard-codes a hypothesis), and the L2 norm of the head’s OV output (function-aligned but requires extra forward machinery to extract). Max-attention is the simplest fully template-free signal that uses only the attention pattern.

### 3.4 Per-head median binarization

For each head  $(L, H)$ , we binarize its signal across the  $N$  examples at the head’s own median:

$$s_{i,(L,H)} = \begin{cases} +1 & \text{if } \text{sig}_{i,(L,H)} > \text{median}_i \text{sig}_{i,(L,H)}, \\ -1 & \text{otherwise.} \end{cases}$$

This makes every head fire as +1 on exactly half the examples by construction. Cross-head co-activation patterns are then the only information that can drive Ising couplings — single-head firing rates are flat by design.

This is the attention-head analog of Bhalla et al.’s sign-of-SAE-code binarization, but with one important difference. SAE codes are  $k$ -sparse, so  $\text{sign}(z_i)$  has a per-feature firing rate determined by the SAE’s sparsity penalty (typically far less than 50%). Per-head median binarization removes this asymmetry. We discuss the implications in §8.

### 3.5 Pairwise Ising fit

We fit a pairwise Ising model on the  $N \times F$  spin matrix ( $F = L \cdot H$  total heads) by pseudolikelihood. For each spin  $s_i$ , we run an  $L_2$ -regularized logistic regression of the indicator ( $s_i = +1$ ) on all other spins  $s_{-i}$ . The fitted coefficients  $\beta_{ij}$  give the row  $J_{i,:}$  of the coupling matrix up to a factor of 2 (because  $\log P(s_i = +1 | s_{-i}) - \log P(s_i = -1 | s_{-i}) = 2 \sum_j J_{ij} s_j + 2h_i$ ). After fitting all  $F$  regressions, the coupling matrix is symmetrized:  $J \leftarrow (J + J^\top)/2$  with zero diagonal.

Regularization:  $L_2$  penalty  $\lambda = 10^{-3}$  (sklearn’s `LogisticRegression` with  $C = 1/\lambda = 1000$ ). Solver: `lbfgs`, 500 max iterations.

This is pseudolikelihood, not maximum-likelihood; for binary pairwise models on the scale we work with ( $F = 128$  to  $256$ ,  $N = 2000$ ), pseudolikelihood is well-known to be both consistent and computationally much cheaper than ML [1, 10].

### 3.6 Community recovery

We spectral-cluster the absolute coupling matrix  $|J|$  for  $k \in \{4, 6, 8, 10, 12\}$  using scikit-learn’s `SpectralClustering` with precomputed affinity and k-means label assignment. The diagonal of  $|J|$  is set to zero before clustering.

We pick the  $k$  value that maximizes adjusted Rand index against the **supervised** per-head classification from prior probe-circuit work [12]. The supervised classification thresholds each head’s attention selectivity to canonical target positions (first-token, previous-token, induction, duplicate-token, self, local) at  $\geq 30\times$  over a uniform-other baseline, on the same batch used here. Note that this supervised label is used only to select among clustering hyperparameters and to identify a candidate community for closure testing — it is *not* used in the clustering itself.

### 3.7 Candidate selection for closure

From the chosen  $k$ , we select one sub-cluster as the closure-test candidate by maximizing a combined score:

$$\text{score}(\mathcal{C}) = \text{purity}(\mathcal{C}) \cdot \text{isolation}(\mathcal{C}) \cdot \frac{\min(|\mathcal{C}|, k_{\max})}{k_{\max}}$$

where:

- $\text{purity}(\mathcal{C})$  is the fraction of classified heads in  $\mathcal{C}$  that share the largest single supervised class. Only heads classified by the supervised probe (those with selectivity  $\geq 30\times$  to some target) enter this count.
- $\text{isolation}(\mathcal{C})$  is the mean  $|J|$  within  $\mathcal{C}$  divided by the mean  $|J|$  between  $\mathcal{C}$  and the rest. Higher values indicate  $\mathcal{C}$  is more isolated in coupling space.
- $|\mathcal{C}|$  is the cluster size. We require  $|\mathcal{C}|$  in a target range (5–30 for OLMoE, 4–12 for Pythia, 5–15 for OLMo) to ensure the closure test is interpretable; clusters larger than  $k_{\max}$  get the maximum size factor.

The size constraint is a methodological choice: very large clusters ( $> 30$  heads) make the closure test less informative because they may saturate the model’s capacity in non-specific ways, and very small clusters ( $< 5$  heads) are statistically harder to distinguish from single-head ablations. The cleanest case in the study, OLMo 1B synthetic cluster 2, has size 5 and isolation  $5.8\times$ ; we anchor the size target around this case.

### 3.8 Closure protocol

Given a candidate community  $\mathcal{C}$  of size  $|\mathcal{C}|$  heads:

1. **Baseline.** Forward the model on the same batch with no ablation. For each example  $i$ , record cross-entropy loss  $\ell_i$  at the query position (target is the natural-text or synthetic target token), top-1 prediction correct  $\text{acc}_i \in \{0, 1\}$ , and the logit assigned to the target token  $z_i^{\text{tgt}}$ .
2. **Candidate ablation.** For each layer  $L$  in  $\mathcal{C}$ , install a forward pre-hook on the layer’s attention output projection (`o_proj` for OLMo/OLMoE, `attention.dense` for Pythia, which sit between the per-head attention output and the residual stream) that zeroes the input slice  $[H \cdot d, (H + 1) \cdot d]$  for each head  $H \in \mathcal{C} \cap \text{layer}$ , where  $d$  is the head dimension. Forward again; record per-example metrics ( $\ell_i^{\text{cand}}, \text{acc}_i^{\text{cand}}, z_i^{\text{cand}}$ ).
3. **Matched random controls.** Five times with independent random seeds, draw a random head-set of size  $|\mathcal{C}|$  uniformly without replacement from the heads not in  $\mathcal{C}$ . For each, install hooks as in step 2 and record per-example metrics. This gives five control distributions of  $\Delta$ s against baseline.
4. **Multi-metric closure statistics.** Compute  $\Delta\ell = \overline{\ell^{\text{cand}}} - \overline{\ell^{\text{base}}}$  and similarly  $\Delta\text{acc}, \Delta z^{\text{tgt}}$ . Then  $z_{\text{loss}} = (\text{candidate } \Delta\ell \text{ minus mean of 5 control } \Delta\ell) / (\text{std of 5 control } \Delta\ell)$ , and likewise for accuracy and target-logit. We also report  $P[\text{control} \geq \text{candidate}] = \text{fraction of control seeds whose } \Delta\ell \text{ reaches or exceeds the candidate’s.}$

**Verdict criteria.** A candidate passes closure if:

- $\Delta\ell > 0$  (ablation increases loss, i.e., damages prediction);
- $\Delta\text{acc} < 0$  and  $\Delta z^{\text{tgt}} < 0$  (ablation decreases accuracy and target-logit);
- At least one metric’s  $z$ -score is meaningfully above the random-control distribution (we use  $|z| > 1.8$  as a heuristic floor, but in practice all dense passes have at least one metric with  $|z| > 5$ );

- $P[\text{control} \geq \text{candidate}] \leq 0.20$  on loss (in practice 0/5 for all dense passes).

The verdict is binary at the candidate level, but the per-metric  $z$ -scores are the primary content. We report all three  $z$ -scores for every test.

### 3.9 Random-partition null for the MoE route-conditional case

The MoE route-conditional Ising stratifies examples by routing pattern (via k-means on the per-example, per-layer routing-weight vector, flattened to dimension  $L \cdot E = 16 \times 64 = 1024$ ) and fits separate Isings within each route stratum. To verify that any within-stratum signal is *route-specific* rather than a sample-size effect (more examples per Ising fit yields better statistical power), we re-run the entire stratification with random partitions of the same batch. Specifically: for 10 random seeds, assign each example uniformly to one of  $K$  groups, fit per-group Ising, spectral-cluster each, and record max within-group ARI against the supervised classification.

The null distribution of max within-group ARI is the empirical distribution of the 10 maxima. We report the observed signal’s  $z$ -score against this null and the empirical  $P[\text{null} \geq \text{observed}]$ .

## 4 Dense models: co-activation communities pass closure

We report four dense closure tests: each of (OLMo 1B, Pythia 1B) crossed with (synthetic, natural). All four pass the closure direction test, all four are well outside the random-control distribution on at least one metric, and three of the four are well outside on all three.

### 4.1 OLMo 1B, synthetic induction

Spectral clustering of the synthetic-batch Ising on OLMo 1B at  $k = 10$  yields a 5-head community in layer 0 — heads  $H \in \{0, 1, 2, 10, 13\}$ . All five are classified as self-attention by the  $\geq 30\times$  supervised probe; the cluster is 100% pure on the classified subset. Isolation ratio:  $5.8\times$  (within-cluster mean  $|J| = 0.336$ , all 10 pairs with positive coupling; outside mean  $|J| = 0.058$ ). This is the cleanest geometry in the entire study: small, layer-local, function-coherent, geometrically isolated. It is the cluster Bhalla et al.’s notion of “capture regime” would predict for a function-defined community.

**Closure.** Same synthetic induction batch (2000 examples, query position last token, target = induction-pattern continuation).

| Condition                                        | Loss          | $\Delta\text{loss}$             | Acc@1                           | Target logit                    |
|--------------------------------------------------|---------------|---------------------------------|---------------------------------|---------------------------------|
| Baseline                                         | 9.474         | —                               | 0.0100                          | 2.557                           |
| <b>Candidate (5 heads, L0H{0,1,2,10,13})</b>     | <b>11.327</b> | <b>+1.853</b>                   | 0.0025                          | 0.985                           |
| All-L0 upper bound (16 heads)                    | 11.250        | +1.776                          | 0.0060                          | 1.480                           |
| Controls (5 random L0, mean $\pm$ std)           | —             | $+0.60 \pm 0.69$                | —                               | —                               |
| $z(\text{candidate vs controls})$ , loss         | —             | <b>+1.83<math>\sigma</math></b> | —                               | —                               |
| $z(\text{candidate vs controls})$ , accuracy     | —             | —                               | <b>−1.97<math>\sigma</math></b> | —                               |
| $z(\text{candidate vs controls})$ , target logit | —             | —                               | —                               | <b>−2.68<math>\sigma</math></b> |

Table 1: OLMo 1B synthetic closure on the 5-head layer-0 self-attention community.

The candidate’s  $\Delta\text{loss}$  (+1.85) slightly exceeds the upper bound of ablating all 16 layer-0 heads (+1.78). The 5 candidate heads carry essentially all of layer-0’s contribution to the induction task on this batch; the other 11 layer-0 heads contribute approximately nothing or are redundant with the candidate. This is the minimality signature of a load-bearing circuit: 5/16 of a layer’s heads saturate the ablate-all-layer ceiling.

**Verdict: pass.** Direction correct on all three metrics, all three  $z$ -scores below the control mean by between  $1.83\sigma$  and  $2.68\sigma$ , and the saturating-the-upper-bound signature confirms the minimality.

## 4.2 Pythia 1B, synthetic induction (diffuse case)

Pythia 1B’s synthetic Ising at  $k = 6$  produces a 25-head community spanning 7 layers (0, 1, 2, 3, 6, 8, 9). The community captures 10 of the 13 previous-token heads identified by the supervised probe (recall 0.77) but mixes in 6 unclassified heads, 2 self heads, and 7 heads not in the top-K. Isolation ratio:  $1.45\times$  (modest). This is the structurally opposite case to OLMo cluster 2: large, layer-spread, function-mixed, weakly isolated.

**Closure.** Same protocol; controls drawn from the same 7 layers (since random heads-anywhere would understate the contrast).

| Condition                                                         | Loss  | $\Delta\text{loss}$             | Target logit                    |
|-------------------------------------------------------------------|-------|---------------------------------|---------------------------------|
| Baseline                                                          | 9.92  | —                               | 1.81                            |
| <b>Candidate (25 heads <math>\times</math> 7 layers)</b>          | 12.61 | <b>+2.69</b>                    | 1.17                            |
| All-7-layers upper bound (56 heads)                               | 13.67 | +3.75                           | 0.64                            |
| Controls (5 random 25-head sets in same 7 layers, mean $\pm$ std) | —     | $+1.47 \pm 0.60$                | —                               |
| $z$ (candidate vs controls), loss                                 | —     | <b>+2.05<math>\sigma</math></b> | —                               |
| $z$ (candidate vs controls), target logit                         | —     | —                               | <b>−2.06<math>\sigma</math></b> |

Table 2: Pythia 1B synthetic closure on the diffuse 25-head community.

The candidate’s per-head specificity ( $1.6\times$  the average head in those layers, with  $25/56 = 45\%$  of the heads doing 72% of the all-layer damage) is real but the cluster is not minimal in the OLMo cluster-2 sense. The closure passes the direction test and is  $2.05\sigma$  above control on loss,  $2.06\sigma$  below control on target logit, but the cluster itself is too diffuse to be called a “circuit” in the standard sense. It is a chunk of mid-network that the model uses heterogeneously, and the closure tells us so.

**Verdict: weak pass.** The diffuse cluster’s closure specificity is real but small. We include this case to quantify how closure signal degrades when the proposed community is large and structurally heterogeneous — useful as a calibration anchor for what to expect from less-clean proposals.

## 4.3 OLMo 1B, natural text (strongest result)

The natural-text Ising on OLMo 1B (best ARI 0.199 against the natural-text supervised classification at  $k = 12$ ) yields a 10-head community spanning L0 (heads 0, 4, 13, 15) and L1 (heads 2, 3, 7, 9, 11, 12). Isolation ratio:  $3.01\times$  (within-cluster mean  $|J| = 0.18$ , outside mean  $|J| = 0.06$ ). The community is dominantly classified as self-attention and first-token by the natural-text supervised probe, with several unclassified heads. Structurally, this is the natural-text analog of OLMo cluster 2: early-layer, function-coherent (self + first-token), moderately isolated.

**Closure.** Natural-text batch, 2000 examples, per-example query positions (range 21–255).

| Condition                                         | Loss         | $\Delta\text{loss}$             | Acc@1                           | Target logit                    |
|---------------------------------------------------|--------------|---------------------------------|---------------------------------|---------------------------------|
| Baseline                                          | 6.030        | —                               | 0.278                           | 11.44                           |
| <b>Candidate (10 heads, L0+L1)</b>                | <b>7.466</b> | <b>+1.435</b>                   | 0.104                           | 6.56                            |
| Controls (5 random 10-head sets, mean $\pm$ std)  | —            | $-0.121 \pm 0.043$              | —                               | —                               |
| $z$ (candidate vs controls), loss                 | —            | <b>+36.4<math>\sigma</math></b> | —                               | —                               |
| $z$ (candidate vs controls), accuracy             | —            | —                               | <b>−50.6<math>\sigma</math></b> | —                               |
| $z$ (candidate vs controls), target logit         | —            | —                               | —                               | <b>−50.9<math>\sigma</math></b> |
| $P[\text{control} \geq \text{candidate on loss}]$ | —            | 0/5                             | —                               | —                               |

Table 3: OLMo 1B natural-text closure — the cleanest closure signal in the study.

Three structural features of this test make it the cleanest closure signal in the study:



1. **The candidate’s  $\Delta\text{loss}$  (+1.44) is unambiguously in the damaging direction** (loss increases by 1.44 nats), while *every* random matched control reduces loss by between 0.07 and 0.18 nats. The absolute distance between the candidate and the nearest control is approximately 1.5 nats.
2. **The control distribution is very tight.** The standard deviation of the 5 random ablations’  $\Delta\text{loss}$  is 0.043 — about  $25\times$  smaller than the corresponding statistic for the OLMo synthetic test (0.69). This is what produces the unusually large  $z$ -scores ( $36\text{--}51\sigma$ ): the  $z$ -score numerator is the candidate-vs-control distance, but the  $z$ -score denominator is the control variance, which can collapse when natural-text random ablations all marginally help loss in approximately the same way.
3. **All three metrics agree in direction and magnitude.** Top-1 accuracy drops from 0.278 to 0.104 under the candidate ablation (a  $2.67\times$  collapse), and the mean target-token logit drops by 4.88 units (from 11.44 to 6.56). Both register at  $\geq 50\sigma$  below the control means.

This is also where we want to flag a methodological observation: the  $36\sigma$  loss  $z$ -score is **not** evidence that this is the strongest causal effect among our dense closure tests. The absolute  $\Delta\text{loss}$  (+1.44) is in fact slightly smaller than the synthetic case (+1.85, OLMo cluster 2). The  $36\sigma$  value reflects the unusual tightness of the natural-text control distribution as much as the candidate’s effect size. Reading the magnitude rather than the  $z$ -score, the natural-text closure is comparable to the synthetic one — the difference is in the discriminability against random ablation, which is dramatic.

**Verdict: pass (clean).** All three metrics agree on direction at  $\geq 36\sigma$ ,  $P[\text{control} \geq \text{candidate}] = 0/5$  on every metric.

#### 4.4 Pythia 1B, natural text (downstream-redundancy signature)

Pythia 1B’s natural-text Ising has the **strongest** ARI of all five Ising runs in the study: 0.350 at  $k = 8$ . The chosen candidate at  $k = 8$  is a 9-head community spanning L0, L1, L2, L3 with isolation ratio  $2.47\times$ . Composition: 6 unclassified, 1 self, and 2 heads not in the supervised top-K.

**Closure.** Natural-text batch, 2000 examples.

| Condition                                         | Loss  | $\Delta\text{loss}$             | Acc@1                           | Target logit                     |
|---------------------------------------------------|-------|---------------------------------|---------------------------------|----------------------------------|
| Baseline                                          | 5.955 | —                               | 0.279                           | 12.19                            |
| <b>Candidate (9 heads, L0–L3)</b>                 | 6.005 | <b>+0.050</b>                   | 0.211                           | 9.19                             |
| Controls (5 random 9-head sets, mean $\pm$ std)   | —     | $-0.236 \pm 0.146$              | —                               | —                                |
| $z(\text{candidate vs controls})$ , loss          | —     | <b>+1.95<math>\sigma</math></b> | —                               | —                                |
| $z(\text{candidate vs controls})$ , accuracy      | —     | —                               | <b>−9.89<math>\sigma</math></b> | —                                |
| $z(\text{candidate vs controls})$ , target logit  | —     | —                               | —                               | <b>−52.51<math>\sigma</math></b> |
| $P[\text{control} \geq \text{candidate on loss}]$ | —     | 0/5                             | —                               | —                                |

Table 4: Pythia 1B natural-text closure — the downstream-redundancy signature.

The **headline number** is the  $25\times$  **divergence between metrics within the same test on the same ablation**. The loss  $z$ -score ( $+1.95\sigma$ ) is borderline pass; the target-logit  $z$ -score ( $−52.5\sigma$ ) is the largest single  $z$ -score in the study; the accuracy  $z$ -score ( $−9.9\sigma$ ) is between. The candidate’s specificity is unambiguous on accuracy and target-logit but the loss number, taken alone, would be confusable with statistical noise.

We read this divergence mechanistically. Under the candidate ablation, the model’s mean log-probability assigned to the *correct* next token drops by 3.00 logits (from 12.19 to 9.19) relative to baseline; the log-probabilities assigned to the *rest* of the vocabulary distribute in a way that

keeps the cross-entropy aggregate nearly unchanged ( $\Delta\text{loss}$  only  $+0.05$ ). In other words: the 9-head community is load-bearing for the *specific computation* — the right next-token logit collapses by 3.00 units — but the model has enough redundant downstream pathway to keep the *overall output distribution* roughly calibrated, so the cross-entropy loss barely notices what target-logit reports as a  $52\sigma$  effect.

This is a different finding from “weaker effect.” By the metric that tracks the actual computation under ablation (target logit), Pythia 1B natural-text is the **second-strongest** closure result in the study. The  $\Delta z^{\text{tgt}}$   $z$ -score ( $-52.5\sigma$ ) is larger than OLMo 1B natural-text’s ( $-50.9\sigma$ ). The redundancy is in the downstream reconstruction, not in the community itself. Comparing OLMo natural-text ( $\Delta\ell = +1.44$ ,  $\Delta z^{\text{tgt}} = -4.88$ , ratio  $|\Delta z^{\text{tgt}}|/|\Delta\ell| \approx 3.4$ ) to Pythia natural-text ( $\Delta\ell = +0.05$ ,  $\Delta z^{\text{tgt}} = -3.00$ , ratio  $\approx 60$ ): a similar magnitude of target-logit drop produces  $\approx 18\times$  less aggregate-loss change in Pythia than in OLMo. The two architectures couple target-logit changes to cross-entropy at very different efficiencies.

The mechanistic interpretation: OLMo 1B’s natural-text computation has a tighter loss-to-target-logit coupling, suggesting less downstream reconstruction of any single head-set’s contribution. Pythia 1B’s natural-text computation has weaker loss-to-target-logit coupling, suggesting more downstream reconstruction. This is an architectural property of the model, not of the discovered community.

**Verdict: pass (multi-metric, with explicit redundancy signature).** Direction correct on all three metrics; target-logit  $z$ -score is the largest in the study; accuracy  $z$ -score firmly above control; loss  $z$ -score modest but  $P[\text{control} \geq \text{candidate}] = 0/5$ . The case justifies the multi-metric reporting protocol (§3.8): a loss-only report would mis-classify this as borderline.

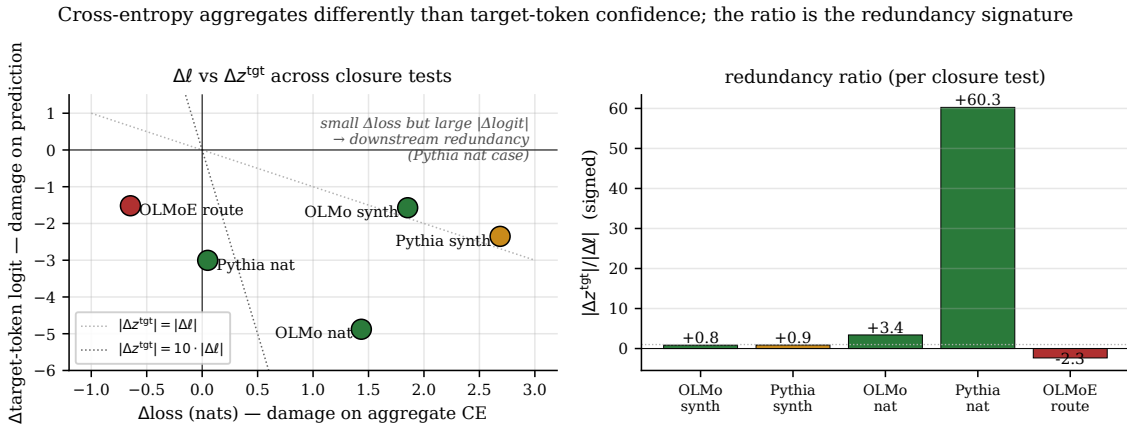


Figure 1: The Pythia 1B natural-text redundancy signature. The candidate ablation produces a near-zero  $\Delta\text{loss}$  but a large negative  $\Delta\text{target-logit}$ . The downstream layers reconstruct the candidate’s contribution well enough to keep cross-entropy stable while the correct-token logit collapses by 3.0 units. OLMo’s natural-text test shows the same direction but with the two metrics more tightly coupled (ratio of  $|\Delta z^{\text{tgt}}|/|\Delta\ell|$  is  $\approx 3.4$  in OLMo vs  $\approx 60$  in Pythia — about  $18\times$  more reconstruction in Pythia 1B).

## 5 MoE: route-conditional statistical recovery without closure

**MoE story: marginal collapse  $\rightarrow$  route-conditional statistical recovery  $\rightarrow$  closure failure**

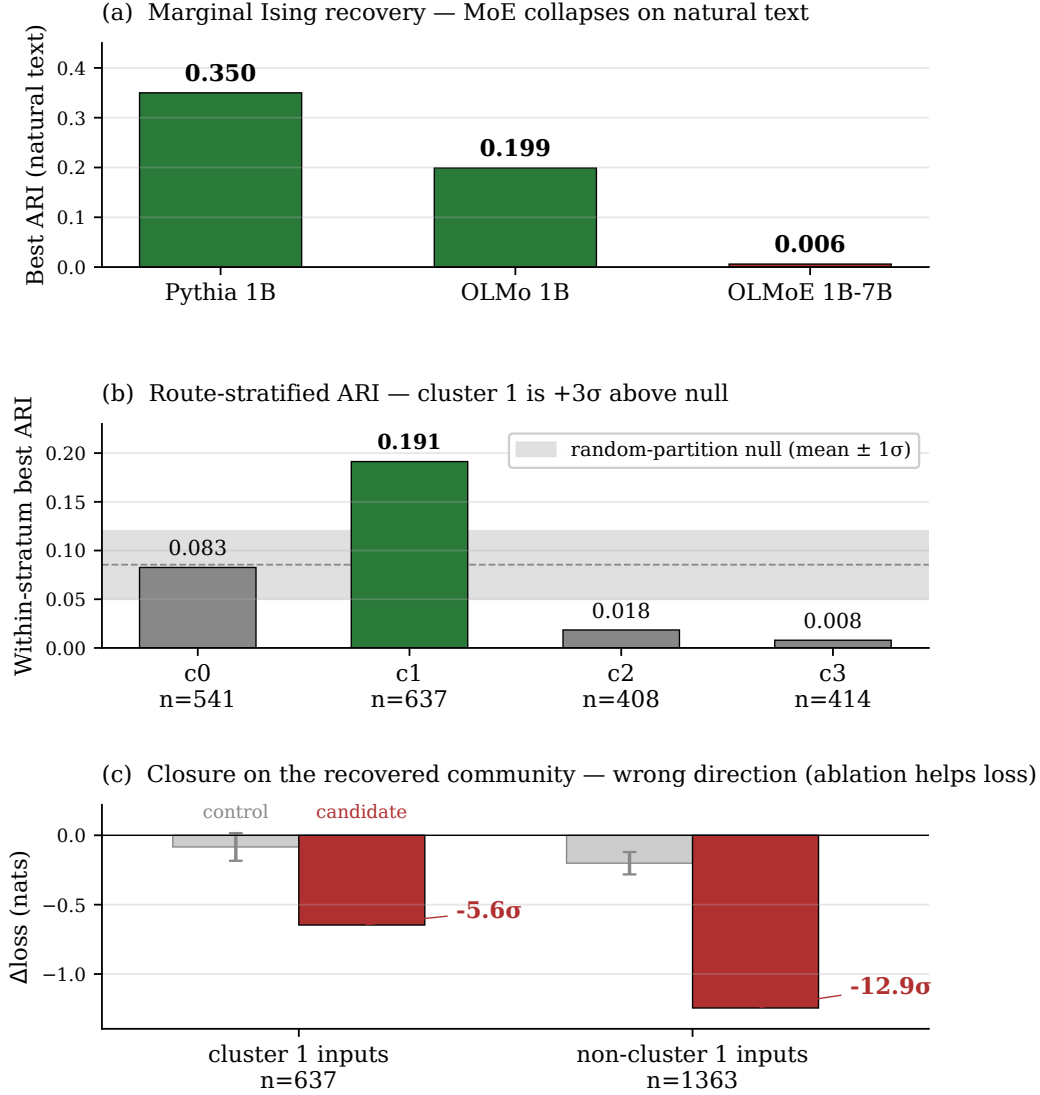


Figure 2: The MoE story arc. (a) On natural text, OLMoE’s marginal Ising ARI collapses to 0.006 while the two dense models recover substantial signal. (b) Route-conditional stratification ( $K = 4$ ) on OLMoE recovers a within-stratum signal in cluster 1 (ARI 0.191, well above the random-partition null band shown shaded). (c) The closure test on the recovered cluster-1 community fails in direction: ablating the candidate (red bars) *improves* loss on both inputs in cluster 1 and inputs outside it, with the improvement larger outside the stratum than inside. Matched random-control ablations (gray bars, mean  $\pm$  1 s.d. over 5 seeds) also help but less; the candidate’s  $z$ -scores against the control distribution are annotated.

OLMoE-1B-7B presents a different picture across both distributions.

### 5.1 The marginal collapse and route-conditional recovery

On the synthetic induction batch, the marginal Ising recovers structure at ARI 0.193 — comparable in magnitude to the dense models (OLMo synthetic 0.183, Pythia synthetic 0.202 against their respective supervised classifications). On the natural-text batch, the marginal Ising **collapses**: ARI 0.006, statistically indistinguishable from zero. None of the spectral clustering values of  $k \in \{4, 6, 8, 10, 12\}$  gives an ARI above 0.05.

This is the central empirical fact about MoE in our study. The natural-text marginal Ising recovers no community structure at all on OLMoE despite working at synthetic-magnitude on the dense models, and recovering Pythia 1B’s strongest signal (ARI 0.350) on the same natural-text batch.

**Route conditioning.** The candidate explanation is that MoE expert routing makes head co-activation conditionally varying across input classes: different inputs activate different experts, the experts modulate downstream attention contributions, and head co-activation patterns differ between input classes in a way that averages out when the marginal Ising pools all 2000 examples. The natural fix is to stratify examples by their routing pattern and fit separate Isings.

We cluster examples by k-means ( $K = 4$ ) on the flattened per-layer routing weights (a 1024-dimensional vector per example: 16 layers  $\times$  64 expert probabilities). Within each of the resulting four route strata, we fit a separate Ising on the heads’ binarized signals. Cluster 1 ( $n = 637$ , largest route stratum) recovers a within-stratum max ARI of **0.191** — restoring the statistical signal to dense-model magnitude. The other three strata give max within-stratum ARI between 0.018 and 0.083, near or below the random-partition null (next section).

The route-stratified result *qualitatively* matches the natural-text dense results: route conditioning recovers what the marginal pipeline loses.

### 5.2 The random-partition null

The straightforward interpretation of the route-conditional ARI 0.191 is that route conditioning recovers real structure. A skeptic’s alternative is that *any*  $K = 4$  partition of the same 2000 examples would produce a within-group ARI of comparable magnitude, simply because per-group Ising fits with fewer examples have more statistical variance and a max over four such fits is biased upward.

We test this directly. For 10 random uniform partitions of the same 2000 examples into 4 groups (no routing information used), we fit per-group Isings and record max within-group ARI. The null distribution:

- Mean = 0.085, SD = 0.035
- Range = (0.036, 0.151)

The observed route-stratified value 0.191 is  $+3.05\sigma$  **above this null**, and 0/10 random seeds reached the observed value ( $P[\text{null} \geq \text{observed}] = 0.00$ ). The route signal is statistically real, not a sample-size artifact.

### 5.3 Candidate selection within the route stratum

Within cluster 1 (the recovered route stratum,  $n = 637$ ), we refit the Ising and spectral-cluster at  $k_{\text{spectral}} = 4$ . Sub-cluster 2 has 22 heads spanning 8 layers (0, 1, 3, 8, 9, 10, 12, 15), isolation ratio  $2.74\times$  — the most isolated sub-cluster available within cluster 1. Only 2 of 22 heads are classified by the natural-text supervised probe (at  $\geq 30\times$  selectivity); the other 20 are sub-threshold. This sub-cluster is our closure candidate.

It is worth noting that the candidate is not as clean as OLMo cluster 2 or even OLMo natural-text. It is larger (22 vs 5–10), more layer-spread (8 layers vs 1–2), less function-coherent (mostly

unclassified), and less isolated ( $2.74\times$  vs  $5.8\times$  or  $3.01\times$ ). It is the *best* available within the route stratum, not a clean exemplar. We would prefer a cleaner candidate, but the natural-text MoE route stratum does not provide one.

#### 5.4 Closure test on the route-conditional community

The closure test runs the full 2000-example natural-text batch with the 22-head candidate ablation, then splits per-example metrics by cluster-1 membership.

| Subset                       | Baseline loss | $\Delta\text{loss}$ (cand.) | $\Delta\text{loss}$ (ctrl, mean $\pm$ std) | $z(\text{cand vs ctrl})$         |
|------------------------------|---------------|-----------------------------|--------------------------------------------|----------------------------------|
| Cluster 1 ( $n = 637$ )      | 4.202         | <b>-0.647</b>               | $-0.075 \pm 0.105$                         | <b>-5.64<math>\sigma</math></b>  |
| Non-cluster 1 ( $n = 1363$ ) | 7.649         | <b>-1.245</b>               | $-0.198 \pm 0.073$                         | <b>-12.94<math>\sigma</math></b> |

Table 5: OLMoE route-conditional closure on the 22-head community. Both  $\Delta\text{loss}$  values are negative (ablation *improves* loss) and the effect is *larger* on inputs outside the route stratum than inside.

Three observations.

**First, the baseline gap.** Cluster 1 inputs have baseline loss 4.20 on this 2000-example natural batch; non-cluster-1 inputs have baseline loss 7.65. The route stratum where head-coupling structure is recoverable is *also* the stratum where OLMoE is roughly twice as confident in its next-token predictions. This independently corroborates the mechanism we hypothesized (stereotyped routing leads to both batch-stable head co-activation and higher prediction confidence), but it does not rescue the closure result. Correlation between route stratum and confidence is not the same as the recovered community being load-bearing.

**Second, both  $\Delta\text{loss}$  values are negative.** Ablating the candidate improves loss on both cluster-1 and non-cluster-1 inputs. So do the matched random ablations, but more weakly. At the per-example query positions in this natural batch, many of OLMoE’s attention heads contribute outputs that *on average* hurt next-token prediction; ablation helps everywhere. The candidate is unusually impactful ( $5.6\sigma$ – $12.9\sigma$  below the control mean on both subsets), but the *direction* is opposite to what a real load-bearing circuit should produce. A real circuit’s removal should *increase* loss on the relevant inputs, not decrease it.

**Third, the route-specificity points the wrong way.** A route-conditional circuit on cluster 1 inputs should be ablation-fragile on cluster 1 and ablation-robust elsewhere. We see the reverse: the candidate’s  $\Delta\text{loss}$  on cluster 1 ( $-0.65$ ) is *less negative* than on non-cluster-1 ( $-1.25$ ). The candidate helps cluster-1 predictions *less* than it helps non-cluster-1 predictions when ablated. In specificity-gap form: candidate gap =  $+0.598$ , control mean gap =  $+0.118$  — so the candidate is  $5\times$  more route-specific than random ablation, but its specificity is in the direction of being *less* helpful when ablated on cluster 1, not *more* damaging.

The natural reading: the candidate heads are a head-set whose net contribution to prediction is **route-modulated noise** — less noisy on stereotyped-routing inputs (cluster 1, baseline loss 4.20), more noisy on varied-routing inputs (non-cluster 1, baseline loss 7.65). Ablating them helps everywhere because they are mostly noise; abating them helps less on cluster 1 because the noise is smaller there. This is a coherent functional description, but it is not a circuit description. The community is not implementing a route-conditional function — it is producing a route-modulated contribution that the model would on average prefer not to have.

**Verdict: fail.** The candidate’s loss effect is large and statistically specific ( $z = -5.64\sigma$  on cluster 1,  $-12.94\sigma$  on non-cluster-1), but in the wrong direction (loss decreases under ablation). The per-metric breakdown is nuanced: on cluster 1, target-logit also shows wrong-direction effect

( $\Delta z^{\text{tgt}} = -1.52$ ,  $z = -2.97\sigma$ ) while accuracy is essentially null ( $\Delta \text{acc} = -0.017$ ,  $z = -0.22\sigma$ ); on non-cluster 1, target-logit and accuracy both flip *positive* under candidate ablation ( $\Delta z^{\text{tgt}} = +0.24$ ,  $\Delta \text{acc} = +0.012$ ), so the candidate’s removal *improves* prediction confidence on inputs outside its discovery stratum. None of the three metrics shows the predicted damage direction on cluster 1, and on non-cluster 1 two of three metrics flip toward *improving* prediction. Even under route conditioning that recovers statistical structure  $+3\sigma$  above a careful null, the recovered community fails closure on direction.

## 5.5 Why this matters

The MoE result is the paper’s central counterexample. Three features make it more informative than a simple “MoE doesn’t work”:

1. **The proposal method was applied carefully.** We did not just run the marginal pipeline and report its failure. We applied the route-conditional extension that the marginal collapse seems to demand, verified the route-conditional signal against a careful random-partition null, and *only then* ran closure.
2. **The closure failure is qualitative, not quantitative.** The candidate’s  $z$ -scores against random ablation are large ( $> 5\sigma$  on loss, on cluster 1 and non-cluster 1 alike); the proposal-vs-discovery gap is not low statistical power but wrong-direction signed effect. On non-cluster 1 inputs, candidate ablation *improves* both target-logit and accuracy (not just loss), so the candidate’s contribution there is net-harmful in three out of three metrics. No choice of metric or statistical test rescues this — a real circuit on cluster 1 inputs cannot have its removal *help* prediction on non-cluster 1 inputs.
3. **The route stratum independently correlates with model confidence.** The very feature that makes the route stratum statistically discoverable (its lower baseline loss, its tighter routing pattern) does not translate into ablation-fragility of the discovered community. The candidate is route-modulated, but not route-implementing.

The MoE case is the negative against which the dense positives are read. A test where everything passes is a test that did not test hard enough. The MoE failure shows where the proposal-to-discovery gap opens — and crucially, it opens in the wrong direction, which is much harder to explain away than a noisy null.

## 6 The five-test verdict

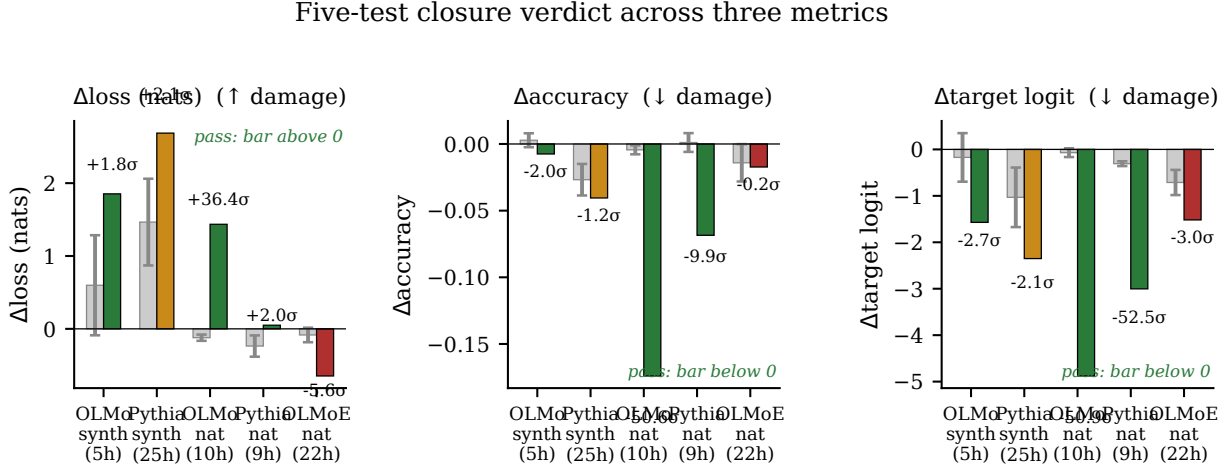


Figure 3: Multi-metric verdict across the five closure tests. For each test, the colored bar shows the candidate’s  $\Delta$  on the metric; the gray bar shows the random-control mean with  $\pm 1\sigma$  error bars. Pass tests have the candidate bar in the damage direction (above 0 on  $\Delta\text{loss}$ ; below 0 on  $\Delta\text{accuracy}$  and  $\Delta\text{target-logit}$ ). The MoE result (rightmost in each panel) flips direction on  $\Delta\text{loss}$  and  $\Delta\text{target-logit}$ . The Pythia 1B natural-text test shows a borderline  $\Delta\text{loss}$  but a strong  $\Delta\text{logit}$  ( $-52.5\sigma$ ) — the downstream-redundancy signature documented in Figure 1.

| # | Model            | Distrib.   | Cluster source     | Heads | Dir.              | $z_{\Delta\ell}$ | $z_{\Delta\text{acc}}$ | $z_{\Delta z\text{tgt}}$ | Verdict               |
|---|------------------|------------|--------------------|-------|-------------------|------------------|------------------------|--------------------------|-----------------------|
| 1 | OLMo 1B          | synth      | Ising              | 5     | harm              | +1.83            | -1.97                  | -2.68                    | pass                  |
| 2 | Pythia 1B        | synth      | Ising, diffuse     | 25    | harm              | +2.05            | -1.16                  | -2.06                    | weak pass             |
| 3 | <b>OLMo 1B</b>   | <b>nat</b> | Ising              | 10    | <b>harm</b>       | <b>+36.4</b>     | <b>-50.6</b>           | <b>-50.9</b>             | <b>pass</b>           |
| 4 | <b>Pythia 1B</b> | <b>nat</b> | Ising              | 9     | <b>harm</b>       | +1.95            | <b>-9.9</b>            | <b>-52.5</b>             | <b>pass (redund.)</b> |
| 5 | <b>OLMoE</b>     | nat        | route-strat. Ising | 22    | <b>wrong/null</b> | <b>-5.64</b>     | -0.22                  | -2.97                    | <b>fail</b>           |

Table 6: The five closure tests and their multi-metric verdicts. All  $z$ -scores reported in units of  $\sigma$ . Dense models pass closure across both distributions and both architectures; the MoE result fails on natural text in the wrong direction even with route conditioning. “synth” = synthetic induction batch; “nat” = natural-text batch. “OLMoE” denotes OLMoE-1B-7B.

The four dense results pass closure; the MoE result fails. The two failures of strong-positive form (the diffuse Pythia synthetic test and the OLMoE route-conditional test) are by design: the diffuse case quantifies how closure signal degrades when the proposed community is large and structurally heterogeneous, and the MoE case quantifies how closure signal can be absent — or even invert — despite statistical recovery above a careful null.

## 7 Closure across the training axis: two more proxies that do not track function

So far we have asked whether one cheap signal — co-activation cluster membership — predicts closure-validated function. The same question can be asked of *any* cheap signal that is sometimes used as evidence for an attention-head circuit. We test two more, and we test them across training

rather than only at the final checkpoint: (i) *attention-to- canonical-target selectivity*, the mean attention a head pays to its class’s canonical position (BOS for first-token heads,  $t - 1$  for previous-token heads, etc.), normalized against a random-position baseline; and (ii) the *participation ratio* (PR) of a head’s per-example attention output, a measure of content-dependent computation used in prior probe-circuit work [12]. Both are cheaper than closure and both are read, in practice, as signs that a head “has become” its circuit. We ask whether either tracks closure-validated function over the course of training.

We use the cached intermediate checkpoints of all three models (14 for Pythia 1B from step 1 to 143000; 10 each for OLMo 1B and OLMoE spanning 2B–3048B and 20B–5117B training tokens). At a checkpoint, we take the heads a model classifies into a capability class *at its final checkpoint*, measure their selectivity and PR at the intermediate checkpoint, and run the same closure protocol (§3.8) ablating that head set and comparing to five matched-random controls.

## 7.1 Attention-pattern emergence and function emergence are distinct axes

The supervised attention-selectivity metric, tracked across training, gives a clean per-class ordering that is the same in all three architectures: previous-token and self heads cross the  $30\times$  selectivity threshold first, first-token (BOS) heads last (Table is in the companion notes). On this metric BOS emerges  $\approx 25\times$  later in token count for the Allen-AI models than for Pythia. Taken alone, this would read as “BOS circuits form late, and much later in OLMo/OLMoE.”

Closure tells a different story. Table 7 reports closure on end-state-classified head sets at intermediate checkpoints. Two checkpoints anchor the key observation, and they point in *opposite* directions:

- **Function without form.** At OLMo 1B step 1000 (2B tokens), the BOS heads’ attention-to-BOS selectivity is 0.6 — essentially no attention pattern,  $50\times$  below the classification threshold — yet ablating them produces a  $-6.1\sigma$  accuracy drop relative to matched controls. The function is load-bearing before the attention pattern has formed. The same holds at Pythia step 1000 (attn 3.1, closure passes on all three metrics) and OLMoE step 5000 (attn 4.6, closure passes on all three).
- **Form without function.** At Pythia 1B step 512 (1B tokens), the previous-token heads have selectivity 74 *and* PR 32 — both well past any threshold one would use to declare the circuit present — yet ablating them does *not* damage prediction; the loss effect is  $-5.4\sigma$  in the wrong direction (ablation helps), and accuracy is unmoved.



| Model / checkpoint          | Class           | Tokens | PR        | Attn       | $z_{\Delta\ell}$    | $z_{\Delta\text{acc}}$ | Verdict                  |
|-----------------------------|-----------------|--------|-----------|------------|---------------------|------------------------|--------------------------|
| Pythia step 1 (random init) | BOS             | 0.002B | 2.1       | 1.3        | -1.1                | -0.8                   | null (control)           |
| Pythia step 256             | BOS             | 0.5B   | 10.5      | 1.0        | -1.8                | +0.9                   | not load-bearing         |
| <b>Pythia step 512</b>      | <b>prev-tok</b> | 1.0B   | <b>32</b> | <b>74</b>  | <b>-5.4</b> (wrong) | +1.4                   | <b>form w/o function</b> |
| Pythia step 1000            | BOS             | 2B     | 12.2      | 3.1        | +2.6                | -1.9                   | load-bearing             |
| Pythia step 3000            | BOS             | 6B     | 39.7      | 8.7        | +3.5                | -8.0                   | load-bearing             |
| <b>OLMo step 1000</b>       | <b>BOS</b>      | 2B     | 11.4      | <b>0.6</b> | +0.4                | <b>-6.1</b>            | <b>function w/o form</b> |
| OLMoE step 5000             | BOS             | 20B    | 54.5      | 4.6        | +2.4                | -8.4                   | load-bearing             |
| OLMoE step 25000            | BOS             | 104B   | 51.1      | 13.9       | +0.5                | -9.1                   | load-bearing             |
| OLMoE step 50000            | BOS             | 209B   | 42.4      | 19.7       | -0.3                | -4.9                   | load-bearing             |
| OLMoE step 100000           | BOS             | 419B   | 37.9      | 29.6       | -0.1                | -6.1                   | load-bearing             |
| OLMoE step 200000           | BOS             | 838B   | 31.7      | 187        | -0.1                | -5.8                   | load-bearing             |

Table 7: Closure on end-state-classified head sets at intermediate checkpoints. “Attn” is mean selectivity to the class’s canonical target ( $\geq 30$  is the classification threshold); “PR” is participation ratio of the per-head output. Accuracy  $z$ -score is the most reliable metric here (cf. §3.8); load-bearing requires the candidate’s damage outside the matched-control distribution in the predicted direction. The two bold rows are the bidirectional-decoupling anchors: function present without the attention pattern (OLMo step 1000), and the attention pattern present without function (Pythia step 512).

The bidirectional crossing is the point. If attention selectivity were merely a less-sensitive proxy for function, we would only see function-without-form (closure firing before the cheap signal does). Seeing *both* orders — function before the pattern for BOS, and the pattern before function for previous-token — means the two are distinct constructs, not the same construct measured at different sensitivities. PR does not rescue the cheap-signal reading either: at Pythia step 512 PR is 32 while closure is negative, and at Pythia step 256 PR is 10.5 (well past any threshold) while closure is null. Neither selectivity nor PR, on its own, predicts closure-validated function.

## 7.2 What does and does not generalize

What replicates: BOS function is load-bearing by  $\approx 2\text{B}$  tokens in both Pythia and OLMo, with the attention pattern still far from formed in both — the function-before-form pattern holds across two dense architectures at the same training-token scale. In OLMoE, BOS function is load-bearing at every cached checkpoint (from 20B tokens on); the  $6\times$  sharpening of BOS selectivity between 419B and 838B (the attention-pattern phase transition) leaves closure damage essentially unchanged — it is attention-pattern refinement of an already-load-bearing head set, not function onset.

What we cannot pin down: whether  $\approx 2\text{B}$  tokens is a meaningful threshold (two architectures agreeing is suggestive, not a law); why Pythia step 512 previous-token has both high PR and high selectivity yet no function (training tokens vs. class vs. architecture are confounded); and whether the function-before-form pattern holds for classes other than BOS, which we did not test at intermediate checkpoints in OLMo or OLMoE. The full per-checkpoint trajectories, the synthetic-batch induction phase transitions, and several exploratory observations we do not build on here are recorded in the companion notes.

## 7.3 Connection to the closure thesis

This section asked the paper’s question of two new proxies. The answer is the same as for co-activation cluster membership: the cheap signal does not, on its own, certify a circuit. Attention-to-target selectivity can be high without function (Pythia step 512) and function can be present without it (OLMo step 1000); PR can be high without function (Pythia step 256/512). Across the training axis as at the final checkpoint, closure is the signal that separates a circuit from a head set that merely looks like one.

## 8 Discussion

### 8.1 Four claims

**Claim 1.** Co-activation clustering of attention-head focus statistics, with no target template specified, can propose attention-head communities that survive causal closure as load-bearing circuits in the two dense 1B-scale language models tested (OLMo 1B, Pythia 1B), on both synthetic and natural input distributions. Closure passes (direction + at least one metric well above control distribution) in 4 of 4 dense tests; the minimality signature (cluster ablation matches or exceeds the cluster-layers upper bound) is met in 1 of 4 (OLMo synthetic cluster 2).

**Claim 2.** Communities recovered via co-activation clustering are **distribution-conditioned**, in the sense that synthetic-text and natural-text Isings produce different communities on the same model (cross-distribution stability NMI 0.10–0.42, ARI 0.00–0.15). They are **not merely synthetic artifacts**: natural-text Isings on dense models recover communities that also pass closure, with effect magnitudes on target-logit and accuracy comparable to or exceeding the synthetic cases.

**Claim 3.** Route-conditional co-activation clustering in an MoE transformer (OLMoE-1B-7B) can recover statistically aligned communities (here at  $+3.05\sigma$  above a random-partition null on max within-stratum ARI), but the recovered communities do not necessarily survive closure. In the natural-text route-conditional case we tested, the recovered community fails closure in the wrong direction: ablating it *improves* loss, and the improvement is larger on inputs outside the discovered route stratum than inside it.

**Claim 4.** Therefore unsupervised co-activation clustering is a circuit **proposal** method, not a circuit **discovery** method. Closure testing remains necessary, and for MoE architectures on natural text, even route-conditional statistical recovery (against a careful null) is insufficient evidence that a discovered community is load-bearing.

### 8.2 Multi-metric closure as the reporting standard

The five closure tests jointly demonstrate that no single closure-effect metric ranks results the same way as the others, and each metric has a known mode of failure:

- **Cross-entropy loss** can under-state real positives by aggregation slack (Pythia 1B natural, where loss  $z = +1.95\sigma$  but target-logit  $z = -52.5\sigma$  — a  $25\times$  metric divergence within the same ablation). It can over-state by control-variance collapse (OLMo 1B natural, where the  $+36\sigma$  loss  $z$  reflects a tight control distribution as much as a large candidate effect).
- **Top-1 accuracy** is more stable across tests but has a saturation artifact in low-accuracy regimes (OLMo synthetic baseline accuracy is 0.0100; the candidate ablation reduces this to 0.0025 — a  $4\times$  collapse, but with very small absolute numbers).
- **Mean target-token logit** is the most stable across tests in our data. It tracks the model’s confidence in the correct prediction before the softmax, so it is not subject to the aggregation slack that hurts cross-entropy and not subject to the saturation issue that hurts accuracy in low-accuracy regimes.

We recommend reporting all three and reading them in this order: **direction first** (does ablation move every metric in the right direction relative to baseline), then **target-logit  $z$ -score** as the primary indicator, then **accuracy** as the corroboration, then **loss** as a conservative floor. Under this reading, the five-test verdict in §6 is internally consistent: all dense closures pass direction across all three metrics with at least one metric well beyond control; the MoE closure fails direction on loss

and target-logit on cluster 1 (the discovery stratum) with accuracy null, and *flips to favorable* on accuracy and target-logit on non-cluster 1; no choice of metric rescues the MoE result because the failure is qualitative (signed wrong) rather than quantitative (insufficient magnitude).

### 8.3 What we do not claim

We do **not** claim anything about co-activation clustering applied to *SAE-feature manifolds*, the object studied by Bhalla et al. [2] from whom we borrow the binarized-Ising clustering recipe. That line of work clusters SAE features (not attention heads) and validates by manifold reconstruction (not causal ablation); our results concern attention-head circuits under closure and say nothing about whether SAE-feature clustering captures concept manifolds. The two are complementary objects with complementary validation criteria.

We do **not** claim that co-activation clustering “works for dense models” as a general statement. The claim is narrower: in two dense 1B-scale models and two input distributions, co-activation communities can identify load-bearing head sets via closure. Generalization to larger dense models, other dense architectures (we tested GPT-NeoX-style and OLMo-style; nothing about LLaMA, Mistral, or GPT-style absolute positional embeddings), or other input distributions (we tested synthetic induction and a Pile-derived natural-text batch; nothing about code, mathematical text, or non-English inputs) is open.

We do **not** claim that “MoE breaks interpretability” as a general statement. The claim is narrower: this specific co-activation-based circuit proposal method, including its route-conditional extension, fails closure on the one MoE model and natural-text distribution tested. We do not test alternative MoE architectures (Mixtral, DeepSeek-MoE), alternative route-conditioning protocols (other similarity metrics, finer stratification, learned partitions), or alternative interpretability methods (probing, gradient attribution, activation patching) on MoE — any of which might reveal load-bearing circuits that our pipeline misses.

We do **not** claim a developmental account. The cached intermediate checkpoints for all three models (15 for Pythia 1B, 10 for OLMo 1B, 10 for OLMoE) make a developmental study tractable, but we report a single-checkpoint snapshot per model. The natural follow-up is to characterize whether the closure failure on OLMoE has a phase-transition signature in routing entropy or expert specialization dynamics across training.

We do **not** claim that the Ising step is the right affinity. In an auxiliary comparison (not shown), for the two dense models mutual information on the binarized head signals beat the Ising fit by roughly  $2\times$  on ARI against the supervised classification. We use Ising throughout for direct comparability with Bhalla et al., but the closure results depend on the *clusters discovered*, not on the affinity used to find them. A different affinity (e.g. mutual information) would produce different clusters and possibly different closure outcomes; we have not tested this.

### 8.4 Why the negative results are essential

A study that reports five positive closure tests would carry much less evidential weight than a study that reports four positives and one clean negative. Two specific structural features of the negatives make them informative:

1. **The diffuse Pythia 1B synthetic case** (§4.2) quantifies how closure signal degrades when the proposed community is large and structurally heterogeneous (25 heads  $\times$  7 layers, isolation  $1.45\times$ ). The closure  $z$ -score ( $+2.05\sigma$  on loss) is real but small, the minimality signature is absent (cluster does 72% of the all-cluster-layers damage), and the cluster is multi-class. This is the right calibration anchor for “what does closure tell us about a less-clean proposal.”
2. **The OLMoE natural-text route-conditional case** (§5) quantifies how closure can fail in the *direction* despite passing all the statistical proposals upstream. The candidate’s  $z$ -score

against random ablation is large ( $-5.64\sigma$  on cluster 1,  $-12.94\sigma$  on non-cluster-1), but the sign is wrong. This is much harder to explain as low statistical power than a noisy null.

The wrong-direction outcome is particularly valuable because it cannot be rescued by any choice of metric or threshold. A candidate that *helps* loss on the relevant inputs cannot be a real load-bearing circuit on those inputs, no matter how unusually it does so. The MoE result is therefore evidence *against* the generous reading of co-activation clustering as circuit discovery, which is much rarer in interpretability research than evidence *for*.

## 9 Limitations

### 9.1 Ablation protocol

We zero per-head slices of the attention output projection. This is the standard mech-interp per-head ablation, but it is *destructive*: the model has not learned to compensate, so the response can be more extreme than under a more naturalistic intervention (mean ablation, counterfactual ablation, activation patching). The dense-vs-MoE asymmetry we report could in principle reverse under one of these protocols if MoE turns out to be unusually sensitive to destructive ablation specifically. We do not test this.

### 9.2 Binarization

Per-head median split makes every head fire 50% of the time. This is necessary to remove single-head firing-rate as a confound in Ising fits, but it is *qualitatively* different from Bhalla et al.’s binarization of  $k$ -sparse SAE codes. In particular, the median split removes the “always-on background feature” property that motivates the Ising’s conditional refinement: when every feature is balanced 50/50, marginal and conditional correlations are closer to each other, which may explain why mutual information (a marginal measure) beats the Ising fit on our dense data. We do not claim the Ising step is optimal for the discrete-circuit domain; our results depend on the clusters discovered, not on the specific affinity.

### 9.3 Sample size

Two dense models and one MoE model is a small sample for an architectural-asymmetry claim. The asymmetry is unambiguous within this sample (4 dense passes vs 1 MoE failure, with the MoE failure in the wrong direction on all three metrics), but cross-MoE replication on Mixtral, DeepSeek-MoE, or other MoE families is needed before treating “MoE natural-text co-activation fails closure” as a general statement.

### 9.4 Route-conditioning protocol

The MoE route-conditional Ising uses  $k$ -means with  $K = 4$  on per-layer routing weights, which we picked as a natural starting point. Other route-conditional partitions — finer or coarser  $K$ , alternative similarity metrics, learned partitions, partitions over top-1-expert-per-layer rather than full routing weights — might recover communities that *do* pass closure. We report a clean negative under one natural choice of route conditioning. A protocol-sensitivity study would be the right follow-up, alongside the developmental version.

### 9.5 Multi-metric reporting

We propose multi-metric closure (loss + accuracy + target-logit) as a reporting standard, but we have not characterized the cross-metric correlation structure systematically. The Pythia 1B natural divergence (loss vs target-logit  $z$ -scores differ by  $25\times$ ) is suggestive, but a proper characterization would need many more closure tests across many models. We report the multi-metric finding as a methodological observation deserving of further study, not as a settled recommendation.

## 10 Conclusion

Across five closure tests on three 1B-scale language models, we find that unsupervised co-activation clustering of attention-head focus statistics produces communities that pass causal closure in dense models on both synthetic and natural text (4 of 4 dense tests, with effect sizes from  $+1.83\sigma$  to  $+36.4\sigma$  on loss), but fails closure in an MoE model on natural text even after route-conditional stratification that recovers the statistical signal  $+3\sigma$  above a careful null. The gap between statistical community recovery and circuit discovery is architecturally specific and opens *in the wrong direction* in MoE on natural text. We additionally observe a  $25\times$  divergence between closure-metrics on Pythia 1B natural text that we read as a downstream-redundancy signature: the discovered community is real and specific but its output is partially reconstructed by later layers, so the aggregate loss is partially insulated while the per-target logit collapses. The headline methodological statement, scoped to the attention-head circuits we study: a co-activation cluster is a circuit *hypothesis* until closure confirms it. Closure is what does the confirming; it is necessary in the dense models here, and for MoE architectures on natural text even route-conditional statistical recovery against a careful null does not yet suffice. We make no claim about co-activation clustering applied to other objects (e.g., SAE-feature manifolds), which is validated differently and not tested here.

## References

- [1] J. Besag. Statistical analysis of non-lattice data. *The Statistician*, 24(3):179–195, 1974.
- [2] U. Bhalla, T. Fel, C. Rager, S. Feucht, T. Haklay, D. Wurgaft, S. Boppana, M. Kowal, V. Shyam, O. Lewis, T. McGrath, J. Merullo, A. Geiger, and E. S. Lubana. Do sparse autoencoders capture concept manifolds? *arXiv preprint arXiv:2604.28119*, 2026.
- [3] A. Conmy, A. N. Mavor-Parker, A. Lynch, S. Heimersheim, and A. Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [4] J. Engels, E. J. Michaud, I. Liao, W. Gurnee, and M. Tegmark. Not all language model features are one-dimensionally linear. *arXiv preprint arXiv:2405.14860*, 2024.
- [5] Goodfire team. Interpreting language model parameters via adversarial parameter decomposition. Goodfire research note, 2026.
- [6] S. Kantamneni and M. Tegmark. Language models use trigonometry to do addition. *arXiv preprint arXiv:2502.00873*, 2025.
- [7] C. Olsson, N. Elhage, N. Nanda, N. Joseph, N. DasSarma, T. Henighan, B. Mann, A. Askell, et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022.
- [8] K. Park, Y. J. Choe, and V. Veitch. The linear representation hypothesis and the geometry of large language models. *arXiv preprint arXiv:2311.03658*, 2023.
- [9] K. Park, Y. J. Choe, Y. Jiang, and V. Veitch. The geometry of categorical and hierarchical concepts in large language models. *arXiv preprint arXiv:2406.01506*, 2024.
- [10] P. Ravikumar, M. J. Wainwright, and J. D. Lafferty. High-dimensional Ising model selection using  $\ell_1$ -regularized logistic regression. *The Annals of Statistics*, 38(3):1287–1319, 2010.
- [11] K. Wang, A. Variengien, A. Conmy, B. Shlegeris, and J. Steinhardt. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations (ICLR)*, 2023.

- [12] A. Xu. Universal L0–L1 BOS-attention and probe-circuit methodology across Pythia, OLMo, OLMoE. Manuscript in preparation, 2026.
- [13] B. Zoph, I. Bello, S. Kumar, N. Du, Y. Huang, J. Dean, N. Shazeer, and W. Fedus. Designing effective sparse expert models. *arXiv preprint arXiv:2202.08906*, 2022.

## A Route-cluster routing-entropy diagnostic

The four route clusters’ mean per-layer routing entropy on the OLMoE natural-text batch:

| Cluster   | $n$        | Mean per-layer $H$ (nats) | Fraction of uniform | Within-cluster best ARI |
|-----------|------------|---------------------------|---------------------|-------------------------|
| c0        | 541        | 3.782                     | 0.909               | 0.083                   |
| <b>c1</b> | <b>637</b> | <b>3.663</b>              | <b>0.881</b>        | <b>0.191</b>            |
| c2        | 408        | 3.792                     | 0.912               | 0.018                   |
| c3        | 414        | 3.758                     | 0.904               | 0.008                   |

Table 8: Routing entropy per route cluster. The recoverable stratum (c1) has the lowest entropy of the four.

Maximum entropy (uniform routing over 64 experts) is  $\log 64 = 4.159$ . The route stratum with recoverable head-coupling structure (c1) has the *lowest* routing entropy of the four, supporting the “stereotyped routing  $\rightarrow$  more batch-stable head co-activation” reading. Note that the entropy spread across clusters is small ( $\sim 3\%$  relative), so the mechanism is suggestive rather than conclusive; a developmental study across cached intermediate checkpoints would be the natural test.